Adjoint-State Gradient for B-Spline Velocity Inversion

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1. Forward Problem

Let $z \in R^{12}$ denote the vector of B-spline control point parameters. The 6 control points are $C_{j(z)} \in R^2$, j = 1,…,6. The B-spline closed curve defines an interface $\Gamma(z)$ in the physical domain $\Omega \subset R^2$. For each spatial point $x \in \Omega$, we define a signed distance $s(x; z)$ to the interface, with $s > 0$ outside the anomaly velocity body and $s < 0$ inside the anomaly body.

The smooth interface velocity model along the B-spline curve is given by:

$$c(x; z) = v_{out} + (v_{in} - v_{out}) * \sigma(-s(x; z)/\tau) \tag{1}$$

where v_{in} and v_{out} are the inside and outside velocities, $\tau > 0$ is a transition width, and $\sigma(\xi)= 1/(1+e^{(-\xi)})$ is the logistic function. That will help make the smooth velocity transition from inside to the outside.

The acoustic wavefield $u(x,t; z)$ satisfies the wave equation:

$$u_{tt} - \nabla\cdot(c(x; z)^2 \nabla u) = \text{source}(x, t) \tag{2}$$

with appropriate boundary and initial conditions.

Synthetic data d is obtained by applying a receiver operator R to the wavefield:

$$d(z) = Ru(z) \tag{3}$$

Assuming Gaussian noise with standard variance σ_d, the data misfit is:

$$J(z) = (1 / (2\sigma_d^2)) \| d(z) - y \|^2 = (1/(2\sigma_d^2)) \| Ru(z) - y \|^2 \tag{4}$$

The ELBO likelihood gradient requires $\nabla_z \log p(y \mid z) = -\nabla_z J(z)$.

2. Lagrangian and Adjoint Equation

To avoid differentiating the full time-stepping procedure, we enforce the wave equation as a constraint via a Lagrange multiplier $\lambda(x,t)$, called the adjoint wavefield. The Lagrangian is:

$$L(z, u, \lambda) = J(z, u) + \int_0^T \int_\Omega \lambda(x,t)(u_{tt} - \nabla\cdot(c(x; z)^2\nabla u) - \text{source}(x,t))\, dx\, dt \tag{5}$$

Variation with respect to u gives the adjoint equation. After integration by parts in time and space, and assuming appropriate boundary and terminal conditions, setting the variation w.r.t. u to zero, the adjoint PDE is:

$$\lambda_{tt} - \nabla\cdot(c(x; z)^2 \nabla\lambda) = (1/\sigma_d^2)\, R^T(Ru - y) \tag{6}$$

with terminal conditions $\lambda(x, T) = 0$ and $\lambda_t(x, T) = 0$, and the equation is solved backward in time.

3. Gradient with Respect to Velocity

Introduce $q(x; z) = c(x; z)^2$. The derivative of the misfit with respect to q is:

$$\partial J/\partial q(x) = \int_0^T \nabla u(x,t) \cdot \nabla \lambda(x,t)\, \mathrm{d}t \tag{7}$$

Since $q = c^2$, we have $\partial q/\partial c = 2c$, so the gradient with respect to $c(x)$ is:

$$\partial J/\partial c(x) = 2c(x) \int_0^T \nabla u(x,t) \cdot \nabla \lambda(x,t)\, \mathrm{d}t \tag{8}$$

This is the standard PDE-level adjoint-state gradient with respect to the velocity field.

4. Chain Rule: Velocity → Signed Distance → Control Points

The velocity depends on the signed distance through the interface law. Differentiating $c(x; z)$ with respect to s gives:

$$\partial c/\partial s(x; z) = (v_{\text{in}} - v_{\text{out}}) * \sigma(1 - \sigma)\,(-1/\tau) \tag{9}$$

Here, $\sigma = \sigma(-s/\tau)$. The B-spline interface curve is parameterized as

$$\mathrm{C}(u) = \sum_{j=1}^{6} N_j(u)\mathrm{C}_j \tag{10}$$

where $N_j(u)$ are B-spline basis functions and C_j are the control points. For each x, let $u^*(x)$ denote the parameter value of the closest point on the curve and $n(x)$ the unit normal. Then the signed distance sensitivity is:

$$\partial \mathrm{s}(x; z)/\partial \mathrm{C}_j = N_j(u^*(x))\, n(x) \tag{11}$$

By the chain rule, the gradient with respect to each control point C_j is:

$$\frac{\partial J}{\partial \mathrm{C}_j} = \int_\Omega \frac{\partial J}{\partial \mathrm{c(x)}} \frac{\partial \mathrm{c(x)}}{\partial \mathrm{s(x)}} \frac{\partial \mathrm{s(x)}}{\partial \mathrm{C}_j}\, dx \tag{12}$$

Substituting the expressions above, we obtain:

$$\frac{\partial J}{\partial \mathrm{C}_j} = \int_\Omega \left[2c(x) \int_0^T \nabla u(x,t) \cdot \nabla \lambda(x,t)\, \mathrm{d}t\right]$$
$$\left[(v_{in-} v_{out})\, \sigma(1 - \sigma)\,(-1/\tau)\right] N_j(u^*(x))\, n(x) \mathrm{d}x \tag{12}$$

Each control point C_j has two components, so this formula yields the full 12-dimensional gradient when all j is assembled.

5. ELBO Likelihood Gradient

Stacking the derivatives with respect to all control points gives $\nabla_z J(z)$. The gradient of the log-likelihood term used in the ELBO is then:

$$\nabla_z \log p(y|z) = -\nabla_z J(z) \tag{13}$$

where $-\nabla_z J(z)$ is obtained by stacking

$$\left(\frac{\partial J}{\partial C_1}, \frac{\partial J}{\partial C_2}, \frac{\partial J}{\partial C_3}, \frac{\partial J}{\partial C_4}, \frac{\partial J}{\partial C_5}, \frac{\partial J}{\partial C_6}\right) \tag{14}$$